

Logical Inference :-

if $P \Rightarrow Q$ and P , then we can infer that Q must be true.

$$\begin{array}{l} \text{I} \dots - P \Rightarrow Q \\ \text{II} \dots - P \\ \hline Q \end{array} \quad \left. \vphantom{\begin{array}{l} \text{I} \dots - P \Rightarrow Q \\ \text{II} \dots - P \\ \hline Q \end{array}} \right\} \text{modus ponens}$$

$$\begin{array}{l} P \Rightarrow Q \\ \sim Q \\ \hline \sim P \end{array} \quad \left. \vphantom{\begin{array}{l} P \Rightarrow Q \\ \sim Q \\ \hline \sim P \end{array}} \right\} \text{modus tollens}$$

$$\begin{array}{l} P \vee Q \\ \sim Q \\ \hline P \end{array} \quad \left. \vphantom{\begin{array}{l} P \vee Q \\ \sim Q \\ \hline P \end{array}} \right\} \text{Elimination}$$

$$\begin{array}{l} P \\ Q \\ \hline P \wedge Q \end{array}$$

$$\frac{P \wedge Q}{P}$$

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$$\rightarrow \begin{array}{l} P \\ \hline Q \vee P \end{array}$$